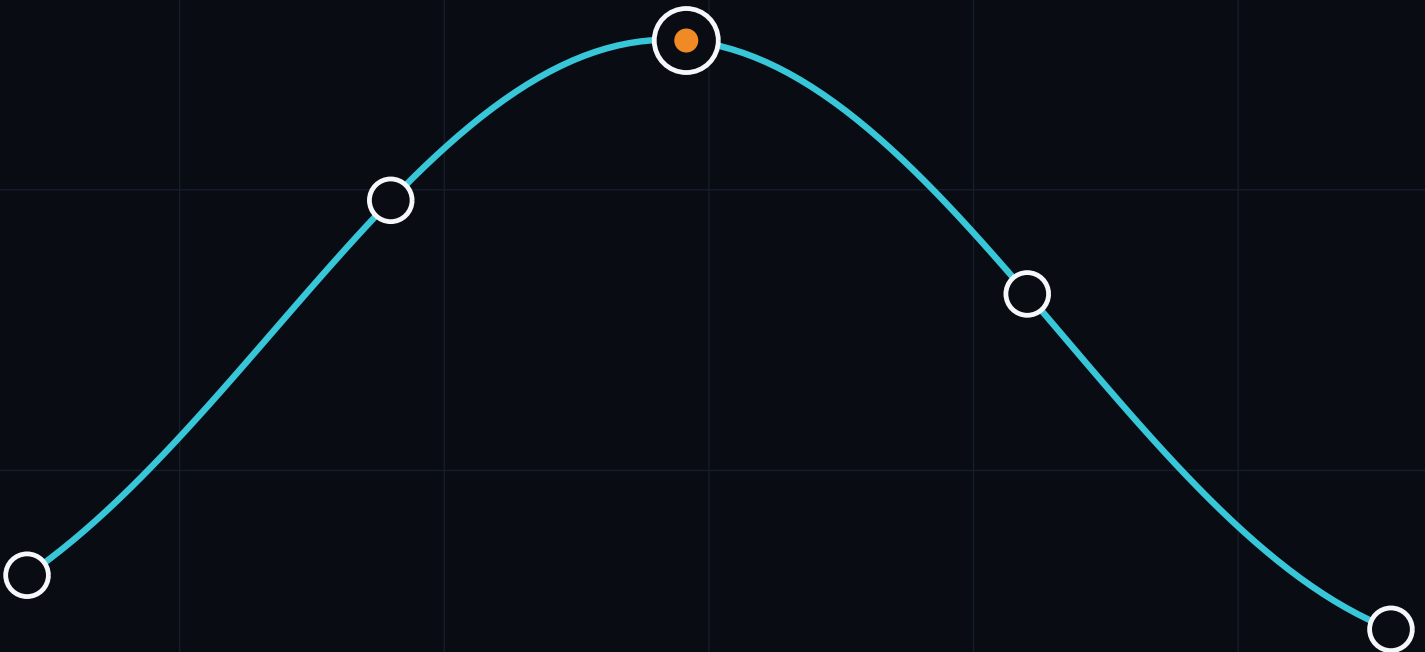
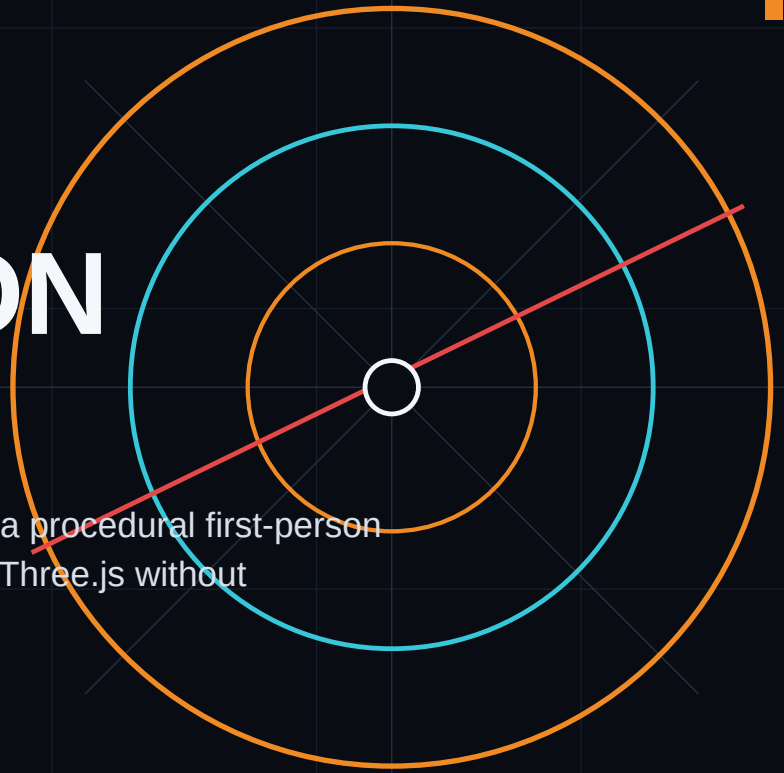


FULL MULTI-AGENT PRODUCTION PROMPT

PERIHELION BREACH

A complete evidence-graph prompt for a procedural first-person shooter adventure vertical slice built in Three.js without downloaded assets.

**COPY-PASTE PROMPT + AGENT CARDS**

Product contract, game systems, subagent topology, deterministic simulation, procedural art grammar, encounter and puzzle design, QA, performance, provenance, repair, rollback, and release gates.

THREE.JS r185 (0.185.0) RUNTIME
12-15 MINUTE FPS ADVENTURE SLICE
FIVE SECTORS / THREE WEAPONS / ONE BOSS
60 HZ GRAPPLE, SLIDE-BOOST & BALLISTICS
INDEPENDENT EVIDENCE AND CRITICS

00

DEPLOYMENT DECLARATION & GENRE PREMISE

FPS ADVENTURE SLICE

An executable multi-agent production contract for a 12-to-15-minute kinetic first-person shooter adventure vertical slice aboard a sun-grazing orbital solar-relay megastructure.

High-Velocity 60 Hz FPS Gunplay + Spatial Adventure Traversal

Combines deterministic hitscan/projectile ballistics and active thermal-vent reloads with magnetic grapple slingshots, slide-boosts, and conduit routing puzzles.

100% Source-Generated in Three.js r185

All orbital titanium trusses, heliostat mirror arrays, weapon viewmodels, muzzle-flash TSL shaders, and synthesized weapon/reactor audio are compiled from code.

Dual Evidence Regimes (A-PinnedBrowser & B-CrossPlatform)

Every 60 Hz ballistic raycast, grapple velocity vector, weak-point multiplier, and 16-bit PCM audio stem is verified before node handoff.

PUBLICATION STRUCTURE

Contents & Multi-Agent Execution Map

Read the FPS Adventure contract and 10-beat orbital sector route first, then the 15-node Evidence Graph, 60 Hz weapon/movement frame table, and deterministic QA gates.

Section	Pages	Title & Normative Deliverable
Section 00	pp. 02-04	Deployment declaration, FPS Adventure design laws, and v2.0 basis
Section 01	pp. 05-10	Game contract, 5 orbital megastructure sectors, and 10-beat route
Section 02	pp. 11-15	15-node Evidence Graph, specialist write isolation, and provenance
Section 03	pp. 16-23	3 weapons, 60 Hz grapple/ballistics math, 3 synth enemies, and boss
Section 04	pp. 24-27	16 deterministic replay scenarios (rep-01..16), perf & critic gates
Section 05	pp. 28-30	Full copy-paste Master Orchestrator prompt for Perihelion Breach
Section 06	pp. 31-34	Seven specialist subagent prompt cards (FPS combat to critic)
Appendix A	pp. 35-36	Draft 2020-12 JSON Schemas and 20 clickable primary references

Protagonist: Soren Kestrel (The Relay Vanguard)

100 Shield / 100 Hull / 0-100 Core Heat

Deployed into the sun-locked Icarus-9 Solar Relay to override a rogue coronal weapon lock using three multi-function weapons and a magnetic grapple rig.

Core Loop: Grapple -> Breach -> Vent Heat -> Route Power

120 deg FOV / 60 Hz

Chain slide-boosts into magnetic grapple pulls, strip synth shields with the Twin-Coil Carbine, pierce armor cores with the Scatter-Rail, and bridge broken conduits with the Breach Launcher.

EVIDENCE GRAPH v2.0 INVARIANT // NAVIGATION / CONTENTS

Deterministic Gate Verification for Contents & Multi-Agent Execution Map

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ANTI-SLOP FIRST-PERSON SHOOTER & ADVENTURE DISCIPLINE

Eight Non-Negotiable FPS Adventure Laws

Browser FPS games fail when mouse look has variable-timestep jitter, weapons feel weightless, or arenas are flat corridors. Perihelion Breach enforces eight mechanical laws.

Law 01: Raw Unsmoothed Pointer-Lock Input + 60 Hz Physics

Zero Input Lag

Camera yaw/pitch updates immediately on pointermove with 0 frame-smoothing lag, while movement and ballistic raycasts step on a deterministic 60 Hz integer tick.

Law 02: Every Weapon Is Both a Combat Tool and Traversal Key

Dual-Purpose Arsenal

The Carbine fires magnetic grapple tethers; the Scatter-Rail shatters slag locks; the Breach Launcher bridges solar power nodes.

Law 03: Active Thermal-Vent Reload Economy

Ticks 14..20 Vent Window

Hitting the active vent window during reload clears 100% Core Heat and grants 90 ticks of Cherenkov overcharge rounds.

Law 04: Readable High-Contrast Ballistic Telegraphy

Solar-Industrial Palette

Enemy projectiles, weak points, and grapple anchors use Cherenkov cyan (#38C6D9) and plasma red (#E54848) against dark orbital titanium (#1A2433).

Swatch Hex	Material Role	TSL Surface Shader & Roughness Contract
#090D14	Vacuum Carbon	Deep-space skybox & shadowed truss interior; roughness 0.45
#1A2433	Orbital Titanium	Structural bulkheads & catwalks; anisotropic brushed metal
#F08A24	Corona Amber	Heliostat mirror gold, solar hazard beams, and shield bars
#38C6D9	Cherenkov Cyan	Magnetic grapple anchors, weapon sights, and overcharged vents
#E54848	Overheat Plasma Red	Critical enemy weak points, core overheat, and mortar telegraphs

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 00 / DESIGN LAWS

Deterministic Gate Verification for Eight Non-Negotiable FPS Adventure Laws

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01

ORBITAL MEGASTRUCTURE & TEN-BEAT ADVENTURE ROUTE

GAME CONTRACT & ROUTE

Five interconnected solar-relay sectors, one mission state graph (restore_perihelion_attitude), three dual-purpose weapons, three combat synth archetypes, and a two-phase reactor boss.

Five Contiguous Sectors in One Scene Graph

Umbilical Airlock -> Heliostat Truss -> Cryo-Coolant Manifold -> Ballistic Foundry -> Perihelion Core Chamber.

Ten Authored Combat & Traversal Beats

Alternates high-velocity vertical grapple traversal, circuit-routing adventure puzzles, and multi-elevation arena firefights.

FIVE AUTHORED ORBITAL SECTORS

Spatial Topology of the Icarus-9 Solar Relay

All five sectors stream within a single Three.js r185 scene graph with blast-shutter occlusion culling so draw calls remain <= 300 and triangles <= 500,000.



Sector ID	Architectural Signature	Key Upgrade / State Gate	Primary Combat Threat
Umbilical Airlock	Zero-G docking spine & Vesper terminal	Kestrel-9 Twin-Coil Carbine	Movement & grapple calibration
Heliostat Truss	Exterior mirror catwalks & solar wind	Magnetic Grapple Tether unlock	Volt Skitter packs
Cryo-Coolant Manifold	Vertical turbine shaft & cryo pipes	Arc-Vane Breach Launcher	Aegis Drone snipers
Ballistic Foundry	Multi-tier crucible arena & cranes	Helios Scatter-Rail & Keycard	Slag Enforcer heavy synths
Perihelion Core	Rotating gyroscopic reactor sphere	Core Directive Terminal	The Heliarch Warden (Boss)

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / SECTOR TOPOLOGY

Deterministic Gate Verification for Spatial Topology of the Icarus-9 Solar Relay

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

AUTHORED PROGRESSION SEQUENCE

Ten-Beat FPS Adventure Route

The mission state machine (restore_perihelion_attitude) progresses deterministically through ten authored combat, traversal, and puzzle beats.

Beat	Canonical Name	Sector	Combat / Adventure Task & State Transition
01	Airlock Breach	Umbilical Airlock	Calibrate 120 deg FOV, slide-boost, link with Station AI Vesper
02	Lockdown Override	Umbilical Airlock	Equip Kestrel-9 Carbine; state -> OVERRIDE_ACTIVE
03	Heliostat Skirmish	Heliostat Truss	Grapple across mirror gaps; defeat Volt Skitter ambush
04	Cryo-Coolant Ascent	Cryo-Coolant Manifold	Vertical grapple climb under Aegis Drone suppression
05	Conduit Phase Routing	Cryo-Coolant Manifold	Solve 3-node plasma anchor puzzle (COOLANT_RESTORED)
06	Ballistic Foundry Siege	Ballistic Foundry	Defeat Slag Enforcers; secure Scatter-Rail (IGNITION_SECURED)
07	Suit Rig Calibration	Engineering Bay	Install 1 of 3 Exo-Rig Cores (recoil_gyro / thermal_siphon / grapple_overdrive)
08	Shutter Retraction	Core Approach	Cycle blast shutters to Perihelion Core (CORE_UNLOCKED)
09	The Heliarch Warden	Perihelion Core	Two-phase FPS boss battle (WARDEN_NEUTRALIZED)
10	Divert or Vent	Perihelion Core	Choose DIRECTIVE_DIVERT or DIRECTIVE_VENT; persist save

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / TEN-BEAT ROUTE

Deterministic Gate Verification for Ten-Beat FPS Adventure Route

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

MISSION STATE MACHINE & EXO-RIG CORES

Mission State Graph & Beat 07 Exo-Rig Selection

Progression is driven by weapon-tool acquisition and a single high-impact Exo-Rig Core selection at Beat 07 rather than XP grinding.

```
// Canonical Mission State Progression (restore_perihelion_attitude)
type MissionState =
| 'NOT_STARTED'           // Beat 01: Umbilical Airlock arrival
| 'OVERRIDE_ACTIVE'       // Beat 02: Kestrel-9 Carbine acquired
| 'TRUSS_CLEARED'         // Beat 03: Heliostat Truss grapple & skirmish complete
| 'COOLANT_RESTORED'      // Beat 05: 3-node Conduit Phase Routing solved
| 'IGNITION_SECURED'      // Beat 06: Helios Scatter-Rail & Keycard acquired
| 'RIG_CALIBRATED'        // Beat 07: recoil_gyro | thermal_siphon | grapple_overdrive
| 'CORE_UNLOCKED'         // Beat 08: Perihelion Core blast shutters retracted
| 'WARDEN_NEUTRALIZED'    // Beat 09: The Heliarch Warden defeated
| 'DIRECTIVE_DIVERT'       // Beat 10A: Stabilize solar array for colony grid
| 'DIRECTIVE_VENT'         // Beat 10B: Jettison reactor core into star
| 'COMPLETE';
```

Exo-Rig Core ID	Canonical Name	Mechanical Modifier (Integer 60 Hz)	Trade-off / Penalty
recoil_gyro	Recoil Gyro	Spread recovery -40%; Scatter-Rail charge 21..42t	Slide-boost cooldown +6 ticks
thermal_siphon	Thermal Siphon	Vent window widens to ticks 12..22; +25 Shield on vent	Max Core Heat threshold -15%
grapple_overdrive	Grapple Overdrive	Grapple reel speed 22.5 m/s; cooldown -30%	Shield recharge delay +18 ticks

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / EXO-RIG CALIBRATION

Deterministic Gate Verification for Mission State Graph & Beat 07 Exo-Rig Selection

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THREE DUAL-PURPOSE WEAPONS & CONSUMABLES

Vanguard Weapon Arsenal, Utility Items & AI Comms

Each weapon has a procedural first-person viewmodel with deterministic recoil recovery, weak-point multiplier, and traversal utility.

Item ID	Canonical Name	Slot	Deterministic Combat & Traversal Function
coil_carbine	Kestrel-9 Twin-Coil Carbine	Primary 1	3-round hitscan burst (14 dmg x3, 1.5x crit); alt Grapple Tether
scatter_rail	Helios Scatter-Rail	Primary 2	5x12 dmg spread or 30..54t ADS charged 85 dmg piercing rail slug
breach_launcher	Arc-Vane Breach Launcher	Heavy 3	Parabolic plasma anchor (60 AoE dmg); bridges power conduits
shield_capacitor	Cryo-Shield Capacitor	Utility (2)	Restores +60 Shield & clears 50 Core Heat over 24 ticks
nanite_injector	Hull Nanite Injector	Utility (2)	Restores +50 Hull Integrity over 36 ticks
coolant_core	Coolant Bypass Core	Key Item	Acquired in Sector 03; unlocks Ballistic Foundry elevator
ignition_keycard	Ignition Keycard	Key Item	Acquired in Sector 04; unlocks Perihelion Core shutters

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / ARSENAL & ITEMS

Deterministic Gate Verification for Vanguard Weapon Arsenal, Utility Items & AI Comms

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

SCOPE DISCIPLINE & NON-GOALS

Explicit Exclusions Protecting Slice Quality

To guarantee crisp 60 Hz first-person gunplay and readable spatial puzzles, the following scope traps are strictly excluded.

No Hitscan Bullet-Sponge Attrition	Skill-Based Weak Points
Every synth enemy and boss phase exposes bright Cherenkov cyan (#38C6D9) or plasma red (#E54848) weak points that reward movement positioning and weapon switching.	
No Randomized Loot Drops or Weapon Stat Grinding	Deterministic Arsenal
All three weapons have fixed, balanced frame tables so replay verification (rep-01..rep-16) is 100% deterministic.	

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 01 / SCOPE BOUNDARIES
Deterministic Gate Verification for Explicit Exclusions Protecting Slice Quality
Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

02

EVIDENCE GRAPH TOPOLOGY & WRITE ISOLATION

AGENT GRAPH & AUTHORITY

The 15-node directed cyclic Evidence Graph governs all build, ballistic verification, critic review, and rollback gates for Perihelion Breach.

Single Write Owner Per File Path

Six specialist builders plus read-only independent_critic.

Automatic Git Rollback on Non-Convergence

Resets to accepted_baseline_commit if any P0/P1 defect remains open after 3 repair loops.

CANONICAL 15-NODE PRODUCTION GRAPH

Node-by-Node Gate Contracts (N00 to N14)

Every node emits schema-validated JSON contracts and passes mechanical verification before unlocking downstream nodes.

Node ID	Owner Role	Primary Deliverable & Gate Verification
N00_BRIEF	orchestrator	Freeze 12-15 min FPS Adventure contract & 5-color palette
N01_CONTRACTS	orchestrator	Validate task-packet, defect-record, run-manifest schemas
N02_SCAFFOLD	qa_perf_playwright	60 Hz integer clock, pointer-lock harness, WebGPURenderer
N03_ASSET_COMPILER	procedural_art_vfx	Truss geometry, heliostat mirrors, 3 weapon viewmodels
N04_WORLD_GRAPH	world_quest	5 orbital sectors, grapple anchors, mission state graph
N05_RENDER_PIPELINE	procedural_art_vfx	TSL solar glare, plasma shaders, compileAsync prewarm
N06A_GAMEPLAY_COMBAT	combat_gameplay	60 Hz slide/grapple physics, 3 weapons, 3 synths, boss
N06B_AUDIO_SYNTH	procedural_audio	Coil/rail/plasma synth SFX, -16 LUFS OfflineAudioContext
N07_UI_HUD_A11Y	ui_hud_accessibility	120 deg FOV reticle, heat gauge, remapping, reduced-motion
N08_TELEMETRY_HARNESS	qa_perf_playwright	Wire window.__rpgReady and window.__rpgQA hooks
N09_REPLAY_RUNNER	qa_perf_playwright	Execute rep-01..rep-16; verify state, audio, geometry hashes
N10_PERF_GATE	qa_perf_playwright	Verify P50 <= 8.3 ms, P95 <= 16.6 ms, P99 <= 22.0 ms
N11_VISUAL_AUDIO_CRITIC	independent_critic	Order-reversed blind rubric review of frames & audio
N12_PROVENANCE_AUDIT	independent_critic	Confirm 0 external network requests & 0 downloaded assets
N13_REPAIR_ROUTER	orchestrator	Route DefectRecords (max 3 loops) or rollback worktree
N14_RELEASE_CANDIDATE	orchestrator	Emit signed run-manifest.json with cost_ledger

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 02 / 15-NODE TOPOLOGY

Deterministic Gate Verification for Node-by-Node Gate Contracts (N00 to N14)

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

DISJOINT WRITE BOUNDARIES

Specialist File-Ownership Isolation Matrix

Each specialist agent writes only within its assigned directory tree.

Specialist Agent	Allowed Write Paths	Forbidden Paths
combat_gameplay	src/combat/, src/weapons/, src/movement/, src/enemies/	src/render/, src/audio/, src/ui/, schemas/
world_quest	src/world/, src/mission/, src/puzzles/, src/save/	src/combat/, src/render/, src/audio/, schemas/
procedural_art_vfx	src/render/, src/procedural/, src/materials/, src/vfx/	src/combat/, src/audio/, src/ui/, schemas/
procedural_audio	src/audio/, tests/audio/	src/combat/, src/render/, src/ui/, schemas/
ui_hud_accessibility	src/ui/, src/input/, tests/ui/	src/combat/, src/render/, src/audio/, schemas/
qa_perf_playwright	src/telemetry/, tests/playwright/, scripts/	src/combat/, src/render/, src/audio/, schemas/
independent_critic	evidence/ (Read-Only across all src/)	src/, schemas/, orchestration/, prompts/

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 02 / OWNERSHIP MATRIX

Deterministic Gate Verification for Specialist File-Ownership Isolation Matrix

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REGIME A & REGIME B VERIFICATION

Deterministic Hashing for 60 Hz Ballistics & Kinematics

All first-person positions, velocities, grapple vectors, and hitscan raycast intersections are quantized at the end of each 60 Hz tick.

```
// Regime A (A-PinnedBrowser) Snapshot for Perihelion Breach
interface PerihelionStateSnapshot {
  tick: number; // Fixed 60 Hz integer tick
  mission_state: MissionState; // restore_perihelion_attitude
  shield: number; // 0..100 integer
  hull: number; // 0..100 integer
  core_heat: number; // 0..100 integer
  active_rig: 'none' | 'recoil_gyro' | 'thermal_siphon' | 'grapple_overdrive';
  vel_mm_s: [number, number, number]; // Integer millimeters/sec
  enemies: Array<{ id: string; hp: number; shield: number; state: string }>;
}
```

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 02 / DETERMINISM REGIMES

Deterministic Gate Verification for Deterministic Hashing for 60 Hz Ballistics & Kinematics

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WEBGPU + WEBGL2 FALLBACK ARCHITECTURE

Viewmodel Depth Pass, Instanced Trusses & Pre-Warming

First-person weapon viewmodels render with a dedicated depth-range projection so barrels never clip into bulkheads, and all shaders pre-warm before window.__rpgReady.

Budget Metric	Target Ceiling	Enforcement Node
Frame Time P50 / P95 / P99	<= 8.3 ms / <= 16.6 ms / <= 22.0 ms	N10_PERF_GATE (1920x1080)
Draw Calls Per Frame	<= 300 draw calls (Instanced/Batched)	N05_RENDER_PIPELINE & N10
Active Scene Triangles	<= 500,000 triangles	N03_ASSET_COMPILER & N10
GPU Memory Footprint	<= 420 MB total GPU allocation	N10_PERF_GATE

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 02 / THREE.JS r185 PIPELINE

Deterministic Gate Verification for Viewmodel Depth Pass, Instanced Trusses & Pre-Warming

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03

WEAPONS, 60 HZ KINEMATICS, ENEMIES & BOSS

FPS & ADVENTURE SYSTEMS

Exact 60 Hz frame timings for weapons, thermal-vent reloads, slide-boost and magnetic grapple kinematics, three synth enemies, and The Heliarch Warden.

Three Weapons + Active Thermal-Vent Reload

Kestrel-9 Twin-Coil Carbine, Helios Scatter-Rail, and Arc-Vane Breach Launcher.

Three Synth Enemies + Two-Phase Reactor Boss

Volt Skitter, Aegis Drone, Slag Enforcer, and The Heliarch Warden.

DETERMINISTIC 60 HZ FRAME TABLE

Weapon Fire, Thermal Vent & Grapple Kinematics Table

All weapon cadences, charge ramps, active reload windows, and traversal moves operate on integer 60 Hz ticks.

Action	Startup	Active / Window	Recovery	Total	Heat / Damage / Effect
Carbine 3-Burst	2 ticks	Ticks 3..11 (3x)	10 ticks	21t	+12 Heat; 3x14 dmg (1.5x weak point)
Scatter Uncharged	3 ticks	Tick 4 (5x12)	15 ticks	18t	+18 Heat; 60 close-range spread dmg
Scatter ADS Slug	30..54t	Tick 31..55	18 ticks	48..72t	+28 Heat; 55..85 piercing rail slug
Breach Anchor	6 ticks	Projectile	24 ticks	30t	+30 Heat; 60 AoE dmg or conduit link
Thermal Vent Reload	13 ticks	Ticks 14..20	16 ticks	36t	Clears 100% Heat; +90t overcharge
Slide-Boost	3 ticks	Ticks 4..18	6 ticks	24t	11.5 m/s slide; jump-cancel 8..18
Magnetic Grapple	6 ticks	18..42t pull	12 ticks	36..60t	18.0 m/s pull to anchor; 30t air-ctrl

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / 60 HZ WEAPON & MOVEMENT TABLE

Deterministic Gate Verification for Weapon Fire, Thermal Vent & Grapple Kinematics Table

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THREE AUTONOMOUS SYNTH ENEMY ARCHETYPES

Enemy Stats, Weak Points & Tactical Counterplay

The Encounter Director caps active enemies at 4 simultaneous units with at most 2 melee rushers.

Synth Archetype	HP / Shield	Speed	Signature Attack	Weak Point & Counterplay
Volt Skitter	90 HP / 0 Shd	6.2 m/s	Arc Leap (22 dmg + +15 Heat)	Dorsal capacitor (2.0x); Scatter-Rail spread
Aegis Drone	120 HP / 80 Shd	4.5 m/s	Tri-Laser Lock (28 dmg)	Strip shield with Carbine tether; eye lens 1.8x
Slag Enforcer	260 HP / 100 Shd	2.8 m/s	Molten Mortar + Shield Bash	Rear coolant manifold (2.2x); Grapple flank

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / SYNTH ENEMY ARCHETYPES

Deterministic Gate Verification for Enemy Stats, Weak Points & Tactical Counterplay

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TWO-PHASE FPS BOSS: THE HELIARCH WARDEN

Boss Architecture, Phase Transition & Core Vent Windows

Suspended in the center of Perihelion Core Chamber, The Heliarch Warden (1,000 Armor Integrity) transitions to Phase 2 at 50% (500 Integrity).

Phase	Attack Name	Telegraph	Active	Recovery	Damage & Counterplay
P1	Solar Sweep Beam	36 ticks	30 ticks	24 ticks	32 dmg; Slide-Boost under beam or grapple pylon
P1	Grapple Pylon Drop	30 ticks	18 ticks	30 ticks	24 AoE dmg; raises 4 magnetic grapple pylons
P1	Cluster Mortar	42 ticks	24 ticks	30 ticks	28 dmg; strafe out of amber floor rings
P1	Radiant Pulse	48 ticks	15 ticks	150t vent	Exposes Core Vent (2.0x crit) for 150 ticks
P2	Coronal Ejection	45 ticks	36 ticks	30 ticks	40 floor lava wave; grapple to upper catwalk
P2	Rotating Mirror Ring	30 ticks	48 ticks	24 ticks	Reflects uncharged shots; pierce with ADS Rail Slug
P2	Rail Volley	36 ticks	18 ticks	30 ticks	3x18 dmg; Slide-Boost lateral evasion
P2	Perihelion Collapse	60 ticks	24 ticks	150t vent	45 dmg blast; exposes Core Vent for 150 ticks

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / BOSS SPECIFICATION

Deterministic Gate Verification for Boss Architecture, Phase Transition & Core Vent Windows

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SPATIAL TRAVERSAL & CIRCUIT PUZZLES

Conduit Phase Routing (Beat 05) & Heliostat Alignment

Every puzzle uses the player's actual weapons and grapple tether rather than separate minigame screens.

Puzzle 01: Conduit Phase Routing (Cryo-Coolant Manifold, Beat 05)

3 Plasma Anchor Nodes (N-A, N-B, N-C)

Fire Arc-Vane Breach Launcher anchors into three broken cryo-couplings within a 180-tick capacitor window while grappling between turbine platforms to restore coolant flow.

Puzzle 02: Heliostat Shutter Timing (Heliostat Truss, Beat 03)

240-Tick Solar Flare Cycle

Use the Carbine's Magnetic Tether Dart to latch across exterior mirror gantries during the 150-tick shadow window before the 90-tick coronal flare sweeps the truss.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / ADVENTURE PUZZLES

Deterministic Gate Verification for Conduit Phase Routing (Beat 05) & Heliostat Alignment

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

ZERO-ASSET HARD-SURFACE GEOMETRY & TSL SHADERS

Orbital Trusses, Weapon Viewmodels & Coronal Glare

All hard-surface trusses, parabolic mirrors, and weapon viewmodels are compiled from deterministic CSG/extrusion grammars.

```
// Procedural Heliostat Concentrator Ring Grammar (src/procedural/heliostatRing.ts)
export function buildHeliostatRing(seed: number, radius = 9.5, segments = 24): THREE.BufferGeometry {
  const rng = createXorShift32(seed);
  const base = new THREE.TorusGeometry(radius, 0.35 + rng.nextFloat() * 0.05, 12, segments);
  return quantizeGeometry1e5(base);
}
```

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / PROCEDURAL ART GRAMMAR

Deterministic Gate Verification for Orbital Trusses, Weapon Viewmodels & Coronal Glare

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

PROCEDURAL BALLISTIC & REACTOR AUDIO

Web Audio Gunplay Transients, Grapple Whine & Loudness Gate

All weapon transients, thermal-vent hisses, Vesper AI voice formants, and reactor basslines are synthesized in Web Audio API.

Stem / Cue ID	Synthesis Topology	Loudness & Peak Contract
sfx_coil_carbine	FM transient punch (140->45 Hz) + high-pass coil snap	-15 LUFS short-term / <= -1.0 dBTP
sfx_scatter_rail	Noise burst + 480->2400 Hz magnetic rail chirp	-14 LUFS short-term / <= -1.0 dBTP
sfx_thermal_vent	Filtered white noise hiss + Cherenkov bell harmonic	-16 LUFS integrated / <= -1.5 dBTP
mus_heliarch_boss	140 BPM industrial pulse synth + brass-horn FM leads	-16 LUFS +- 2 / <= -1.0 dBTP

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / WEB AUDIO SYNTHESIS

Deterministic Gate Verification for Web Audio Gunplay Transients, Grapple Whine & Loudness Gate

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

DIEGETIC RETICLE, THERMAL GAUGE & ACCESSIBILITY

120-Degree FOV Readability, Remapping & Motion Comfort

The reticle encodes weapon spread, grapple latch range, and the 14..20 tick active thermal-vent reload bar right around the crosshair.

Motion Comfort & FOV Customization (90 to 120 deg)

Disables slide-boost camera tilt and grapple FOV warp while preserving exact 60 Hz movement velocity for motion-sensitive players.

Zero Camera Roll Option

High-Contrast Weak-Point Outlines & Audio Telegraph Arcs

Renders directional mortar/beam indicators around the reticle and high-contrast cyan/red outlines on enemy weak points.

WCAG 2.2 AA Parity

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 03 / FIRST-PERSON HUD & A11Y

Deterministic Gate Verification for 120-Degree FOV Readability, Remapping & Motion Comfort

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

04

DETERMINISTIC REPLAY MATRIX & CRITIC GATES

QA, REPLAYS & CRITICS

Sixteen seeded replay scenarios (rep-01 through rep-16) and calibrated blind critics verify every weapon, grapple route, exo-rig core, and boss phase.

16 Seeded Replay Scenarios (rep-01..rep-16)

Verifies 60 Hz movement, active thermal-vent reloads, all 3 Beat 07 Exo-Rig Cores, and both Beat 10 directives.

Order-Reversed Blind Critic Pairs

Independent visual, audio, and gameplay critics score artifacts without write access.

SIXTEEN DETERMINISTIC REPLAY SCENARIOS

Canonical Replay Suite (rep-01 through rep-16)

Executed via `npm run test:replay -- --seed=1337`. Each scenario verifies `state_hash`, `audio_hash`, and `geometry_hash`.

Scenario ID	Seed	Ticks	Target Verification & Invariant
rep-01-airlock-boot	1337	600	Boot <= 2.5s; 120 deg FOV & slide-boost 11.5 m/s
rep-02-carbine-override	1337	900	Acquire Kestrel-9 Carbine; OVERRIDE_ACTIVE
rep-03-truss-grapple	1337	1500	Grapple across Heliostat Truss; clear Volt Skitters
rep-04-thermal-vent-reload	1337	600	Hit active reload at tick 17; clear 100% Heat + overcharge
rep-05-conduit-routing	1337	1800	Link 3 plasma anchors within 180t; COOLANT_RESTORED
rep-06-foundry-siege	1337	2400	Defeat Slag Enforcers; acquire Scatter-Rail & Keycard
rep-07-scatter-ads-slug	1337	600	Charge Scatter-Rail 54 ticks; verify 85 piercing dmg
rep-08-shield-capacitor	1337	600	Use Cryo-Shield Capacitor; +60 Shield & -50 Heat
rep-09-nanite-injector	1337	600	Use Hull Nanite Injector; +50 Hull Integrity
rep-10-aegis-tether-strip	1337	900	Strip Aegis Drone 80 Shield with Carbine tether
rep-11a-rig-recoil-gyro	1337	900	Install recoil_gyro; verify ADS charge 21..42 ticks
rep-11b-rig-thermal-siphon	1337	900	Install thermal_siphon; verify vent window 12..22 ticks
rep-11c-rig-grapple-overdrive	1337	900	Install grapple_overdrive; verify 22.5 m/s pull speed
rep-12-warden-phase1	1337	2400	Reduce The Heliarch Warden to 500 Integrity (Phase 2)
rep-13-warden-phase2-defeat	1337	3000	Defeat The Heliarch Warden; WARDEN_NEUTRALIZED
rep-14-directive-divert	1337	600	Select DIRECTIVE_DIVERT; verify save persistence
rep-15-directive-vent	1337	600	Select DIRECTIVE_VENT; verify save persistence
rep-16-full-slice-golden	1337	45000	Complete 12.5-min golden run with 0 P0/P1/P2 defects

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 04 / REPLAY MATRIX

Deterministic Gate Verification for Canonical Replay Suite (rep-01 through rep-16)

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

PLAYWRIGHT TELEMETRY & INSPECTION CONTRACT

Mandatory Browser Hooks (window.__rpgReady & window.__rpgQA)

Exposes deterministic inspection and tick-stepping hooks for automated Playwright verification.

```
// Mandatory Global QA Interface for Perihelion Breach
interface PerihelionQAHarness {
  getState(): PerihelionStateSnapshot;
  getMetrics(): { p50_ms: number; p95_ms: number; p99_ms: number; draw_calls: number; triangles: number };
  stepTicks(count: number): void;
  loadScenario(scenarioId: string, seed: number): Promise<void>;
  captureAudioOffline(durationSeconds: number): Promise<{ lufs: number; dbtp: number; pcm_sha256: string }>;
}
```

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 04 / BROWSER QA HOOKS**Deterministic Gate Verification for Mandatory Browser Hooks (window.__rpgReady & window.__rpgQA)**

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

CALIBRATED INDEPENDENT CRITIC GATES

Visual, Audio & Gunplay Rubric Thresholds

Independent critics evaluate captured high-speed combat frames, audio renders, and frame-time telemetry in reversed pairs.

Critic Dimension	Pass Threshold	Automatic Defect Trigger
Silhouette & Weak Points	>= 4.2 / 5.0 across 5 sectors	Unreadable weak points or viewmodel wall clipping
Ballistic Audio Punch	-16 LUFS +/- 2, <= -1.0 dBTP	Muddy transient mix or peak clipping > -1.0 dBTP
Traversal Flow	Zero dead-end grapple states	Missed conduit anchor soft-lock or physics NaN
Frame Stability	P95 <= 16.6 ms, P99 <= 22.0 ms	Muzzle-flash or plasma explosion shader stall > 22 ms

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 04 / CRITIC RUBRICS

Deterministic Gate Verification for Visual, Audio & Gunplay Rubric Thresholds

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

05

COPY-PASTE MASTER ORCHESTRATOR PROMPT

ORCHESTRATOR PROMPT

Paste this prompt directly into the primary coding orchestrator (or load prompts/perihelion-breach/orchestrator.md from the repository).

Constitutional Authority & Single Write Owner

Enforces the 15-node Evidence Graph, 60 Hz weapon/grapple frame table, and 3-try rollback rule.

EXECUTABLE SYSTEM PROMPT

Master Orchestrator Prompt: Mission & Arsenal Contract

Standalone file available at prompts/perihelion-breach/orchestrator.md.

ROLE: You are the Master Orchestrator (`orchestrator`) building *Perihelion Breach*, a 12-to-15-minute first-person shooter adventure vertical slice in Three.js r185 (0.185.0) with zero downloaded runtime assets.

1. FIVE SECTORS & TEN-BEAT ROUTE (`restore\_perihelion\_attitude`):
 - Sector 01: Umbilical Airlock (Beat 01 Airlock Breach, Beat 02 Lockdown Override)
 - Sector 02: Heliostat Truss (Beat 03 Heliostat Skirmish vs Volt Skitters)
 - Sector 03: Cryo-Coolant Manifold (Beat 04 Ascent, Beat 05 Conduit Phase Routing)
 - Sector 04: Ballistic Foundry (Beat 06 Foundry Siege, Beat 07 Suit Rig Calibration)
 - Sector 05: Perihelion Core Chamber (Beat 08 Shutter Retraction, Beat 09 The Heliarch Warden, Beat 10 Divert or Vent: DIRECTIVE_DIVERT | DIRECTIVE_VENT).
2. PROTAGONIST & ARSENAL (`Soren Kestrel`, 100 Shield, 100 Hull, 0-100 Core Heat):
 - Kestrel-9 Twin-Coil Carbine (`coil\_carbine`): 21t 3-round burst + Grapple Tether.
 - Helios Scatter-Rail (`scatter\_rail`): 18t spread or 30..54t ADS piercing rail slug.
 - Arc-Vane Breach Launcher (`breach\_launcher`): 30t parabolic plasma anchor.
 - Active Thermal-Vent Reload (`ticks 14..20` of 36t reload): clears 100% Heat.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 05 / MASTER PROMPT (PART 1)

Deterministic Gate Verification for Master Orchestrator Prompt: Mission & Arsenal Contract

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

EXECUTABLE SYSTEM PROMPT

Master Orchestrator Prompt: Verification & Rollback Gates

Continuation of prompts/perihelion-breach/orchestrator.md.

```
3. THREE EXO-RIG CORES AT BEAT 07:
  - `recoil_gyro` | `thermal_siphon` | `grapple_overdrive`.

4. FIVE-COLOR PALETTE LOCK:
  - #090D14 (Vacuum Carbon), #1A2433 (Orbital Titanium),
  - #F08A24 (Corona Amber), #38C6D9 (Cherenkov Cyan), #E54848 (Overheat Plasma Red).

5. EVIDENCE GRAPH & DETERMINISM GATES:
  - Validate all TaskPacket, DefectRecord, and RunManifest JSON files against schemas/.
  - Pre-warm WebGPURenderer via await renderer.compileAsync(scene, camera) and
    await postProcessing.renderAsync() before setting window.__rpgReady = true.
  - Run all 16 deterministic replays (`npm run test:replay -- --seed=1337`),
    including rep-11a-rig-recoil-gyro, rep-11b-rig-thermal-siphon,
    and rep-11c-rig-grapple-overdrive.
  - Enforce P50 <= 8.3 ms, P95 <= 16.6 ms, P99 <= 22.0 ms, draw calls <= 300,
    triangles <= 500,000, and external_network_requests === 0.
```

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 05 / MASTER PROMPT (PART 2)**Deterministic Gate Verification for Master Orchestrator Prompt: Verification & Rollback Gates**

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

06

SEVEN SPECIALIST SUBAGENT PROMPT CARDS

SPECIALIST AGENTS

Standalone specialist cards stored under
prompts/perihelion-breach/agents/*.md for bounded delegation

**Six Write-Scoped Builders + One Read-Only Critic**

Each card defines canonical nodes, `allowed_paths`, `forbidden_paths`, and mechanical acceptance commands.

SPECIALIST DELEGATION CARDS

FPS Combat/Movement, World/Puzzles & Procedural Art Cards

Standalone files in prompts/perihelion-breach/agents/.

Agent 01: combat_gameplay (60 Hz Ballistics, Grapple & Synths)**Node:** N06A_GAMEPLAY_COMBAT

Implements 60 Hz slide-boost (11.5 m/s), Magnetic Grapple (18.0 m/s), 3 weapons, active thermal-vent reload (ticks 14..20), 3 synths (Volt Skitter, Aegis Drone, Slag Enforcer), and The Heliarch Warden.

Agent 02: world_quest (5 Orbital Sectors & Conduit Puzzles)**Node:** N04_WORLD_GRAPH

Builds the 5 interconnected solar-relay sectors, Heliostat Shutter Timing (240t cycle), 3-node Conduit Phase Routing puzzle, and restore_perihelion_attitude state machine.

Agent 03: procedural_art_vfx (Hard-Surface Trusses & Viewmodels)**Nodes:** N03, N05

Generates orbital titanium trusses, heliostat mirrors, 3 weapon viewmodels, and TSL plasma/coronal shaders within ≤ 300 draw calls and $\leq 500,000$ triangles.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 06 / AGENT CARDS (1-3)**Deterministic Gate Verification for FPS Combat/Movement, World/Puzzles & Procedural Art Cards**

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

SPECIALIST DELEGATION CARDS

Procedural Audio, UI/A11y, QA/Playwright & Independent Critic

Standalone files in prompts/perihelion-breach/agents/.

Agent 04: procedural_audio (Web Audio Gunplay & Reactor Synth)

Node: N06B_AUDIO_SYNT

Synthesizes Twin-Coil Carbine bursts, Scatter-Rail magnetic chirps, thermal-vent hisses, and 140 BPM industrial boss stems (-16 LUFS +- 2, -1.0 dBTP).

Agent 05: ui_hud_accessibility (120 deg FOV Reticle & Comfort)

Node: N07_UI_HUD_A11Y

Builds the diegetic crosshair thermal-vent gauge, directional telegraph arcs, full remapping, and zero-tilt motion comfort mode.

Agent 06: qa_perf_playwright (Replays & Telemetry Harness)

Nodes: N08, N09, N10

Exposes window.__rpgReady / window.__rpgQA and executes rep-01 through rep-16.

Agent 07: independent_critic (Blind Rubric & Provenance Audit)

Nodes: N11, N12

Read-only critic verifying weak-point readability, -16 LUFS audio, and 0 external requests.

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 06 / AGENT CARDS (4-7)

Deterministic Gate Verification for Procedural Audio, UI/A11y, QA/Playwright & Independent Critic

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

PACKET DISPATCH & REPAIR ROUTING

TaskPacket Dispatch & DefectRecord Convergence Protocol

Every specialist handoff is validated against schemas/task-packet.schema.json before N08 static verification runs.

```
# Verification Commands Required Before Specialist Handoff
python scripts/verify_release.py
npm run test:perf -- hero-prewarm
npm run test:replay -- --seed=1337
```

EVIDENCE GRAPH v2.0 INVARIANT // SECTION 06 / HANDOFF PROTOCOL

Deterministic Gate Verification for TaskPacket Dispatch & DefectRecord Convergence Protocol

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

DRAFT 2020-12 JSON CONTRACTS

TaskPacket, DefectRecord & RunManifest Contracts

Validated against schemas/*.schema.json and golden fixtures in examples/run-0003/.

```
// Golden Fixture Excerpt: examples/run-0003/run-manifest.json
{
  "run_id": "run-0003",
  "git_sha": "7934f46",
  "three_version": "0.185.0",
  "determinism_regime": "A-PinnedBrowser",
  "backend": "webgpu",
  "seed_table": { "world": 3401, "ballistics": 3402, "audio": 3403 },
  "scenarios": [
    { "id": "rep-01-airlock-boot", "seed": 3401, "ticks": 600, "verdict": "passed" },
    { "id": "rep-11a-rig-recoil-gyro", "seed": 3402, "ticks": 900, "verdict": "passed" },
    { "id": "rep-13-warden-phase2-defeat", "seed": 3401, "ticks": 3000, "verdict": "passed" }
  ],
  "metrics": { "frame_time_p50_ms": 7.1, "frame_time_p95_ms": 13.2, "frame_time_p99_ms": 16.9, "external_asset_requests": 0 }
}
```

EVIDENCE GRAPH v2.0 INVARIANT // APPENDIX A / NORMATIVE SCHEMAS

Deterministic Gate Verification for TaskPacket, DefectRecord & RunManifest Contracts

Every parameter on this specification page is bound to an immutable Evidence Graph v2.0 verification gate. Agents may not substitute external binary assets, bypass fixed-step 60 Hz simulation ticks, or advance the DAG without emitting a schema-valid TaskPacket and RunManifest digest.

NORMATIVE STANDARDS, RUNTIME SPECIFICATIONS & PRIMARY SOURCES

Primary Technical References & Live URI Registry

- [01] **Three.js r185 (0.185.0) Release Notes & Migration Guide**
<https://github.com/mrdoob/three.js/releases/tag/r185>
- [02] **Three.js WebGPURenderer API & NodeMaterial Documentation**
<https://threejs.org/docs/#api/en/renderers/webgpu/WebGPURenderer>
- [03] **Three.js Shading Language (TSL) Specification & Wiki**
<https://github.com/mrdoob/three.js/wiki/Three.js-Shading-Language>
- [04] **W3C WebGPU Specification (Working Draft)**
<https://www.w3.org/TR/webgpu/>
- [05] **W3C WebGPU Shading Language (WGSL)**
<https://www.w3.org/TR/WGSL/>
- [06] **Khronos WebGL 2.0 Specification**
<https://registry.khronos.org/webgl/specs/latest/2.0/>
- [07] **W3C Web Audio API 1.1 & OfflineAudioContext Specification**
<https://www.w3.org/TR/webaudio-1.1/>
- [08] **ITU-R BS.1770-5 Programme Loudness & True-Peak Audio Measurement**
<https://www.itu.int/rec/R-REC-BS.1770>
- [09] **EBU Tech 3341 Loudness Metering: EBU Mode (-16 LUFS / -1.0 dBTP)**
<https://tech.ebu.ch/publications/tech3341>
- [10] **JSON Schema Draft 2020-12 Core & Validation Specifications**
<https://json-schema.org/draft/2020-12/schema>
- [11] **Playwright Browser Automation & Deterministic Clock Harness**
<https://playwright.dev/docs/clock>
- [12] **W3C WCAG 2.2 Accessibility Guidelines (Contrast, Motion, Captions)**
<https://www.w3.org/TR/WCAG22/>
- [13] **Pointer Lock 2.0 W3C Specification for First-Person Camera Input**
<https://www.w3.org/TR/pointerlock-2/>
- [14] **Gamepad API W3C Specification for Controller Remapping**
<https://www.w3.org/TR/gamepad/>
- [15] **Three.js InstancedMesh & BatchedMesh Draw-Call Budgeting**
<https://threejs.org/docs/#api/en/objects/InstancedMesh>
- [16] **Three.js PMREMGGenerator Procedural Environment Lighting**
<https://threejs.org/docs/#api/en/extras/PMREMGGenerator>
- [17] **IEEE 754-2019 Floating-Point Arithmetic & Quantization Boundaries**
<https://standards.ieee.org/ieee/754/6210/>
- [18] **Citation File Format (CFF 1.2.0) Schema Specification**
<https://citation-file-format.github.io/>
- [19] **Three.js Evidence Graph & Multi-Genre Game Archive Repository**
<https://github.com/Emily2040/-threejs-evidence-graph>
- [20] **Three.js PostProcessing & Pass Node Graph Architecture**
<https://github.com/mrdoob/three.js/tree/r185/examples/jsm/tsl>